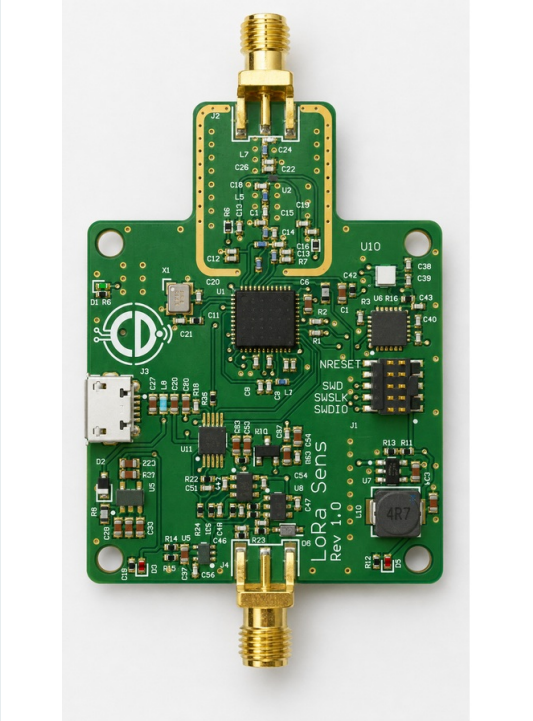
ENGINEERING PORTFOLIO

PCB and embedded electronics engineering

Hardware design in Altium Designer, supported by circuit simulation, embedded firmware and manufacturing documentation.



STM32WL board visualisation. Render, not a bench-test photograph.

Hardware architecture through PCB implementation

I develop circuit architecture, schematics and multilayer layouts, with particular attention to analogue acquisition, RF paths, power distribution and practical test access.

Design decisions supported by evidence

Altium constraints and DRC, LTspice models, Python signal analysis and manufacturing files show how I check the design before hardware testing.

Firmware and connected applications

Nordic BLE and ESP32-S3 demonstrations add sensor acquisition, wireless telemetry and live client integration on physical development hardware.

Project	Distinct engineering evidence
STM32WL LoRa sensor node	Altium; analogue front end; 14-bit ADC; RF matching; LTspice and Python analysis
USB3 to Gigabit Ethernet	Altium layer stack, impedance profiles, differential-pair and length constraints, DRC
Nordic nRF52840	Custom PCB; Zephyr/NCS; embedded C; custom BLE GATT; live EnviroVue client
ESP32-S3	Four-layer PCB; embedded C++; Wi-Fi/HTTP/JSON; physical sensor telemetry
EdgeLink S3 Cellular	LTE-M/NB-IoT/GNSS integration; charger and power path; modem interfaces; assembly package

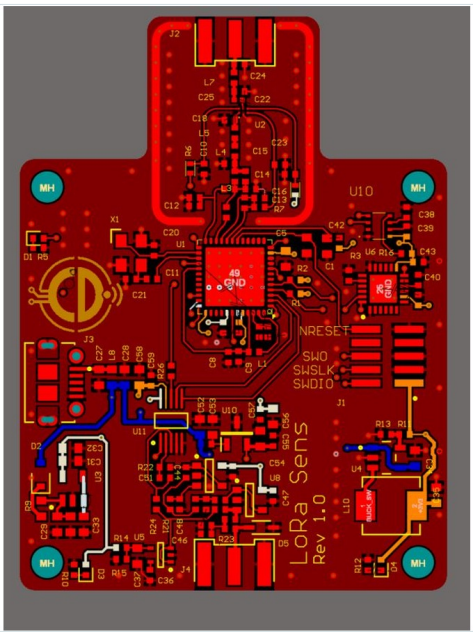
Tools: Altium Designer; LTspice; Python/Jupyter; Zephyr and nRF Connect SDK; PlatformIO. KiCad supports the Nordic, ESP32-S3 and cellular projects.

Sources: STM32WLE5 Rev 2.0 report; USB Altium screenshots; Nordic v14; ESP32-S3 and EdgeLink portfolios.

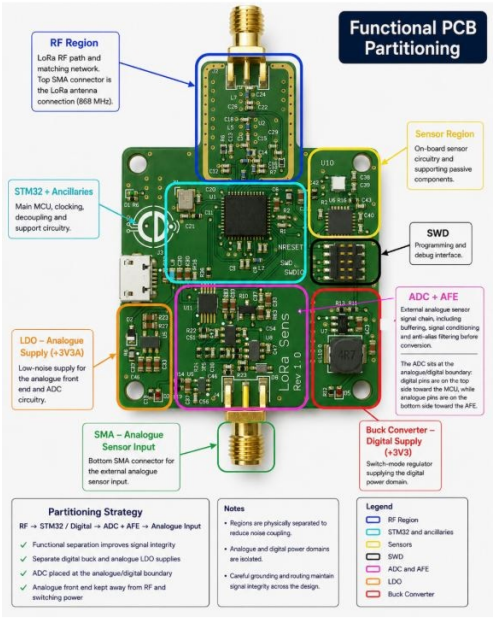
LEAD HARDWARE CASE STUDY / ALTIUM DESIGNER

STM32WL mixed-signal and RF design

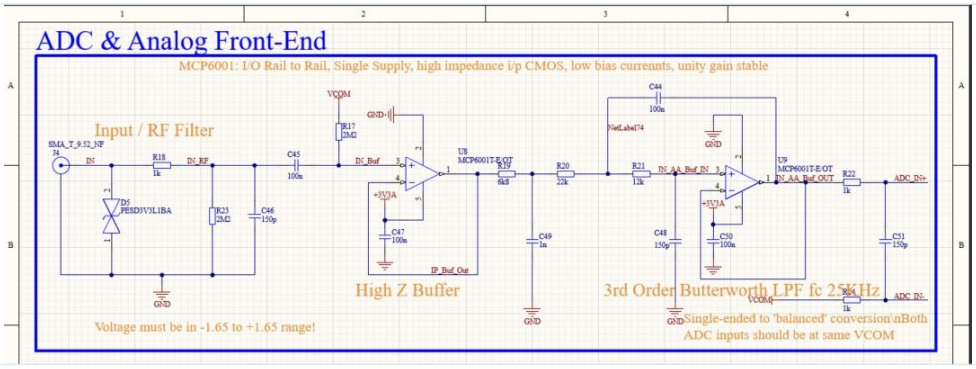
A four-layer sensor node combining an analogue acquisition chain, precision conversion, separate supply domains and an 868 MHz radio interface.



Altium copper layout: RF at the antenna end; MCU centrally; analogue and power circuitry below.



Functional partitioning of RF, MCU, sensing, ADC/AFE and separate supply domains. Annotated board visualisation.



Protected SMA input, high-impedance buffer and active anti-alias filter from the Altium schematic.

Analogue acquisition and reference

MCP6001 buffering and a third-order low-pass filter precede the ADC141S626 14-bit SPI ADC. REF3033 supplies the 3.3 V reference; VCOM provides the signal bias.

Power, RF and PCB implementation

TLV62569 digital supply and LD3985 analogue LDO share a continuous ground reference. BGS12SN6 switching, matching/filtering, a dedicated RF net class and via fencing support the RF section.

Verification: schematic and layout completed; analogue and RF simulations documented; Altium DRC snapshot reports 0 warnings and 0 violations. Power, ADC, sensor and RF bench verification is planned.

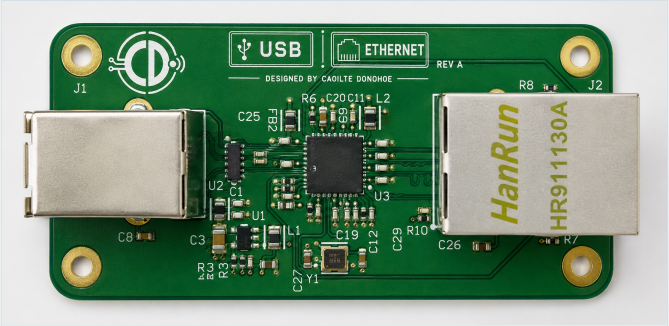
Detailed STM32WLE5 hardware report

Source: STM32WLE5 Hardware Design Report Rev 2.0, sections 2-8 and 11; CAD and rendered visuals.

HIGH-SPEED HARDWARE / ALTIUM DESIGNER

USB3 to Gigabit Ethernet

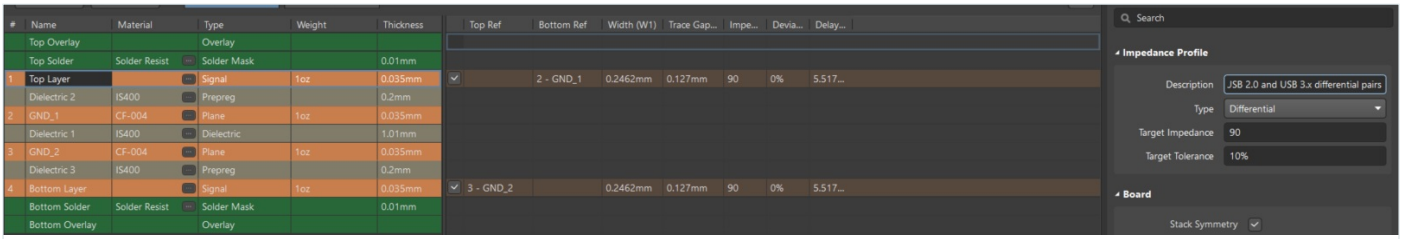
LAN7800 four-layer design with stack-up-defined differential routing and explicit rules for USB, Ethernet, power and clock nets.



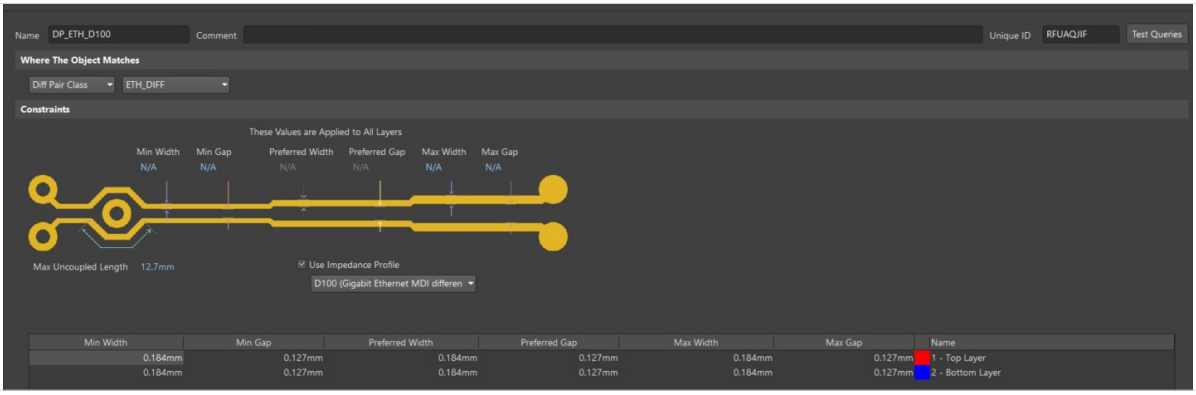
LAN7800 board visualisation from the existing project portfolio.

Constraint-driven layout

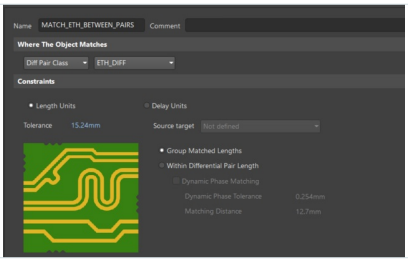
USB uses a 90 ohm differential profile. Ethernet has a separate D100 profile and ETH_DIFF routing scope. Named net classes separate USB2_HS, USB3_SS, ETH_MDI, clock, power and switch nodes.



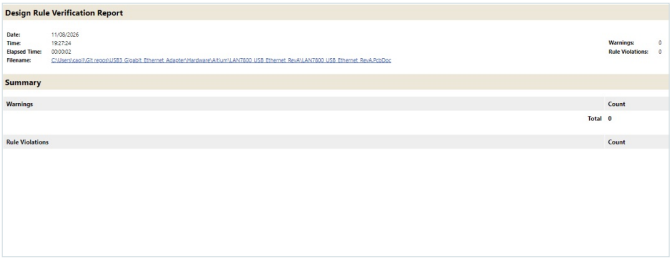
USB profile in Altium: 90 ohm target, 10% tolerance, 0.2462 mm trace width and 0.127 mm pair gap in this snapshot.



Ethernet differential-pair rule linked to its impedance profile; layer-specific width and gap values are explicit.



Matched-length rule scoped to ETH_DIFF. A separate rule controls within-pair matching.



USB/Ethernet DRC report snapshot: 0 warnings and 0 rule violations for the configured checks.

Engineering evidence: the layer-stack snapshot uses outer signal layers and two internal ground planes. Routing rules connect the electrical targets to PCB geometry. DRC results document the configured checks; they do not establish measured USB or Ethernet performance.

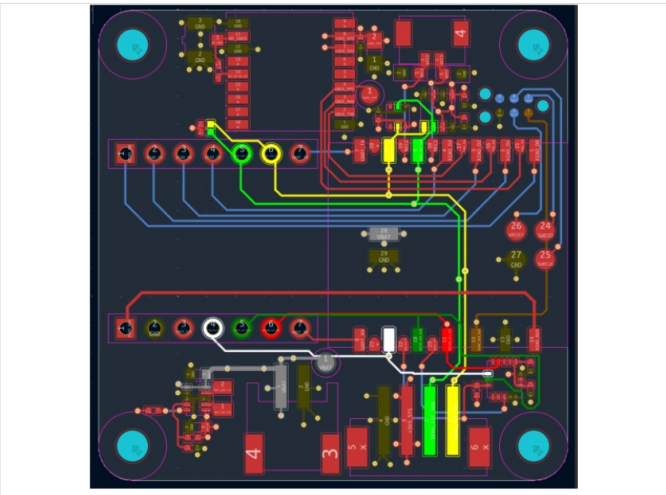
[USB3-Gigabit Ethernet design repository]

Sources: supplied USB Altium stack, impedance, net-class, differential and length-rule screenshots; DRC dated 11 Aug 2026.

WIRELESS FIRMWARE AND HARDWARE

Nordic BLE and Zephyr integration

Custom nRF52840 hardware design paired with embedded C, nRF Connect SDK and a demonstrated sensor-to-browser BLE data path.



Custom Nordic PCB layout in KiCad: GNSS, power, sensor and expansion interfaces.

```
CCC_CALLBACK(humidity)
CCC_CALLBACK(pressure)
CCC_CALLBACK(gas)
CCC_CALLBACK(dew)
CCC_CALLBACK(record)
CCC_CALLBACK(light)

#define MEASUREMENT(uuid, index, callback) \
    BT_GATT_CHARACTERISTIC(uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY, \
        BT_GATT_PERM_READ, read_measurement, NULL, (void *)&measurement_indices[index]), \
    BT_GATT_CCC(callback, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)

BT_GATT_SERVICE_DEFINE(environmental_service,
    BT_GATT_PRIMARY_SERVICE(&service_uuid),
    MEASUREMENT(&temperature_uuid.uuid, 0, temperature_ccc),
    MEASUREMENT(&humidity_uuid.uuid, 1, humidity_ccc),
    MEASUREMENT(&pressure_uuid.uuid, 2, pressure_ccc),
    MEASUREMENT(&gas_uuid.uuid, 3, gas_ccc),
    MEASUREMENT(&dew_uuid.uuid, 4, dew_ccc),
    BT_GATT_CHARACTERISTIC(&record_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_record, NULL, NULL),
    BT_GATT_CCC(record_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE),
    BT_GATT_CHARACTERISTIC(&interval_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_WRITE,
        BT_GATT_PERM_READ | BT_GATT_PERM_WRITE, read_interval, write_interval, NULL),
    BT_GATT_CHARACTERISTIC(&light_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_light, NULL, NULL),
    BT_GATT_CCC(light_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)
);

static const struct bt_data advertising[] = {
    BT_DATA_BYTES(BT_DATA_FLAGS, BT_LE_AD_GENERAL | BT_LE_AD_NO_BREDR),
    BT_DATA_BYTES(BT_DATA_UUID128_ALL, ENV_UUID(0x8f3a0000)),
};
```

Zephyr custom GATT implementation: READ/NOTIFY, CCC subscriptions and writable configuration.

Hardware design

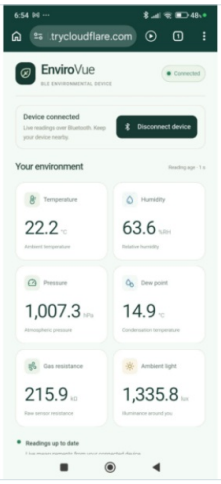
MAX-M10S GNSS and active-antenna interface; NFC connections; TPS63802 battery regulation; sensor integration, SWD and modular I2C/SPI/UART expansion.

Firmware implementation

Modular sensor, BLE, display and application code; Zephyr configuration and logging; persistent settings; custom 128-bit service and writable update interval.



Physical Nordic development setup with live OLED sensor values and BLE connection state.



EnviroVue client showing live measurements received over BLE.

Evidence scope: custom PCB design and manufacturing preparation are separate from the physical development-board firmware demonstration. NFC/GNSS interface design is not presented as an operational NFC/GNSS test.

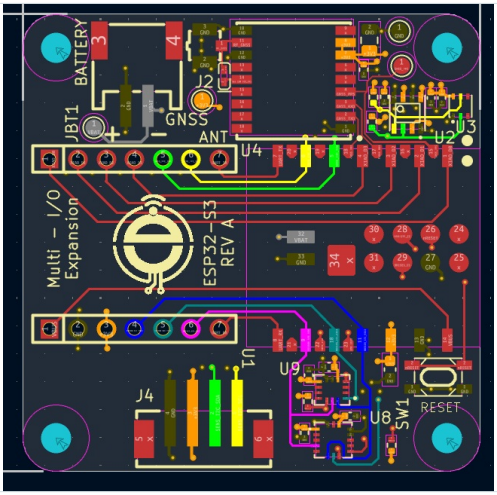
Nordic BLE engineering portfolio v14

Source: Nordic BLE Engineering Portfolio v14, pages 3-9; firmware testing on separate Nordic development hardware.

CONNECTED SENSING AND APPLICATION INTEGRATION

ESP32-S3 Wi-Fi telemetry

A compact four-layer hardware design with modular embedded C++, physical sensor acquisition and a live Wi-Fi/JSON client.



Custom 41 x 41 mm PCB layout in KiCad, including GNSS, sensing and expansion.

```
void sendJson(int code, const String &json) {
  server.sendHeader("Access-Control-Allow-Origin", "*");
  server.sendHeader("Access-Control-Allow-Methods", "GET, OPTIONS");
  server.sendHeader("Access-Control-Allow-Headers", "Content-Type");
  server.send(code, "application/json", json);
}

String statusJson() {
  const bool connected = WiFi.status() == WL_CONNECTED;
  String json = "{\"device\":\"SenseLink X ESP32-S3 Rev A\",\"sensor_setup\":\"development\",\"uptime_ms\
  json += String(millis());
  json += ",\"wifi_configured\":";
  json += credentialsConfigured ? "true" : "false";
  json += ",\"wifi_connected\":";
  json += connected ? "true" : "false";
  json += ",\"wifi_rssi_dbm\":";
  json += connected ? String(WiFi.RSSI()) : String("null");
  json += ",\"ip_address\":";
  json += connected ? String("\"") + WiFi.localIP().toString() + "\"" : String("null");
  json += ",\"last_sample_uptime_ms\":" + String(currentState->uptimeMs);
  json += "}";
  return json;
}

void connectWifi() {
  lastAttemptMs = millis();
  WiFi.begin(SENSELINK_WIFI_SSID, SENSELINK_WIFI_PASSWORD);
  Serial.println("Wi-Fi connection attempt started.");
}
}
```

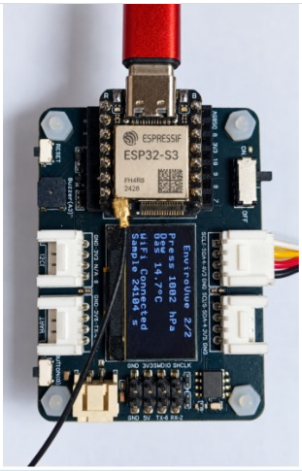
Wi-Fi connection handling and structured status/HTTP response implementation.

Hardware integration

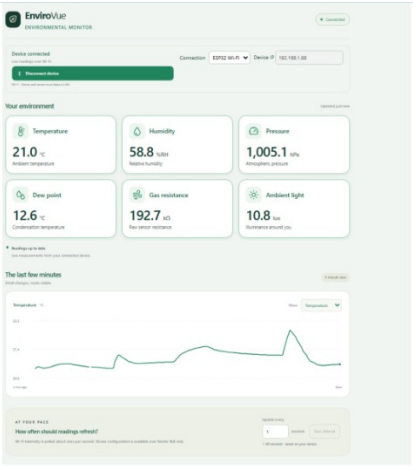
MAX-M10S GNSS, BMI323 IMU, ADXL372 high-g sensing, BME688 and VEML6035 interfaces, battery power and accessible peripheral expansion.

Embedded C++ and connectivity

PlatformIO/Arduino application with separate communications, display, sensor-state and environmental-maths modules. JSON carries sensor values, device status and heartbeat.



Physical ESP32-S3 development hardware with sensors and OLED output.



EnviroVue Wi-Fi client showing environmental values and recent trends.

Evidence scope: firmware and application demonstrations used development hardware with BME680 and BH1750 sensors. They are separate from custom-PCB assembly testing. Wi-Fi is demonstrated; BLE firmware operation is not asserted.

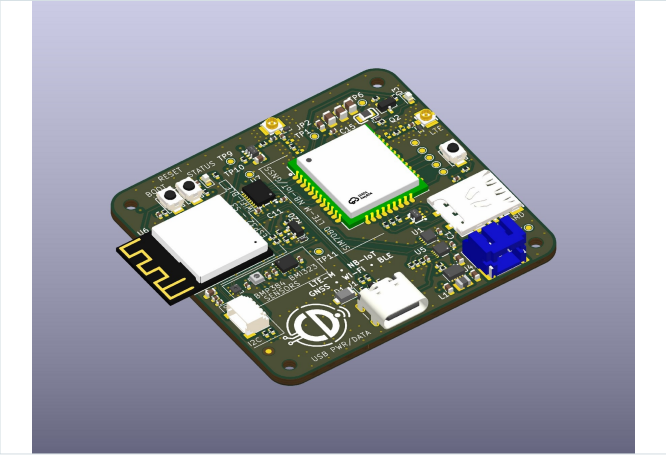
ESP32-S3 hardware and embedded engineering portfolio

Source: ESP32-S3 Engineering Portfolio, pages 3-9; design, manufacturing preparation and development-platform testing.

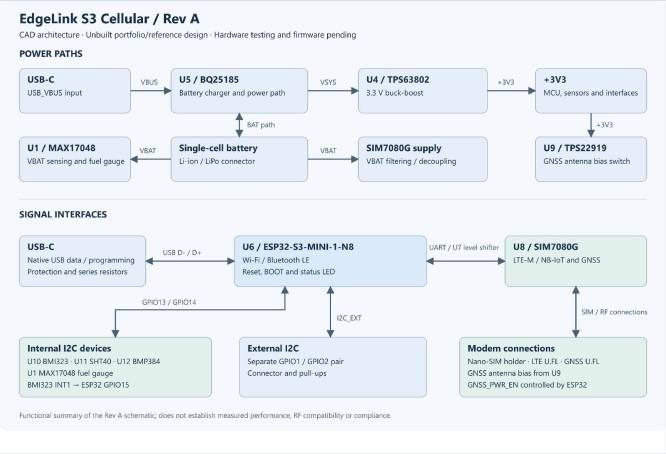
CELLULAR HARDWARE AND POWER ARCHITECTURE

EdgeLink S3 Cellular

ESP32-S3 and SIM7080G integration on a 56 x 52 mm four-layer board, with USB-C, battery power, GNSS and digital sensing.



Rev A KiCad board render. CAD design, not a fabricated prototype.



Power and interface architecture linking the MCU, modem, sensors and battery system.

Charging, system power and modem supply

USB-C feeds a BQ25185 charger and power-path circuit. TPS63802 generates the 3.3 V logic rail; the modem uses a filtered battery supply. MAX17048 supports fuel gauging, with switched GNSS antenna bias.

Digital interfaces and wireless integration

Native ESP32-S3 USB, a level-shifted modem UART, Nano-SIM and LTE/GNSS antenna interfaces. Internal I2C connects BMI323, SHT40, BMP384 and the fuel gauge; external I2C remains separately accessible.

Placement and manufacturing documentation

Ground and supply layers support subsystem placement and routing. Editable KiCad sources sit alongside four-layer Gerbers, drill data, assembly BOM, CPL and readable architecture/schematic documents.

Documented handoff	Evidence
Assembly BOM	46 component groups with LCSC sourcing references
Placement data	89 fitted references, all on the top side
Fabrication and source	Four-layer Gerbers, drill files and editable KiCad project

Current status: Rev A CAD and manufacturing preparation completed. Fabrication, electrical bring-up, modem firmware, RF measurements and compliance testing remain pending.

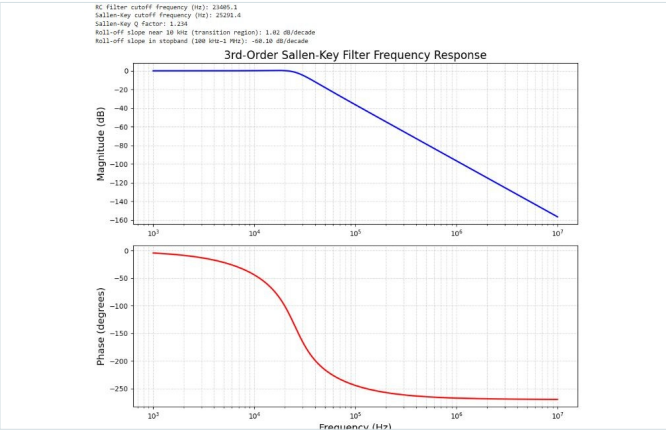
EdgeLink S3 Cellular engineering portfolio

Source: EdgeLink S3 Cellular Engineering Portfolio, pages 1-6; manufacturing counts as recorded in the report.

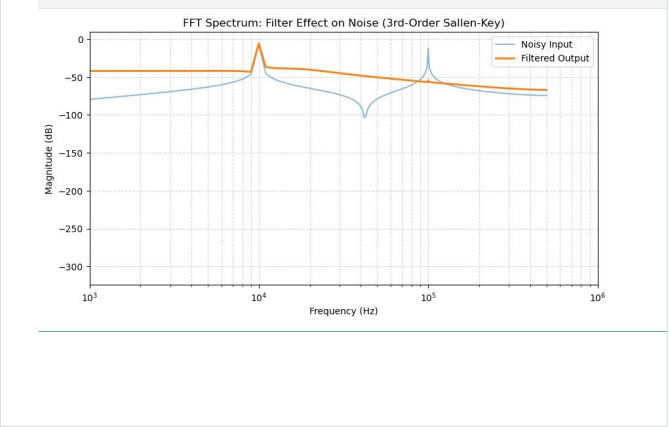
SIMULATION, VERIFICATION AND MANUFACTURING

Evidence behind the design decisions

STM32WL analysis leads this page; Altium DRC and assembly-preparation evidence complement the circuit and signal models.



STM32WL Python magnitude/phase analysis of the analogue low-pass network.



FFT test: wanted 10 kHz component retained; injected 100 kHz content attenuated.

Analogue and RF simulation

LTspice AC and transient checks were compared with Python/NumPy/SciPy results. RF models explored the 868 MHz match and the trade-off between output filtering, passband loss and second-harmonic rejection.



Design Rule Verification Report

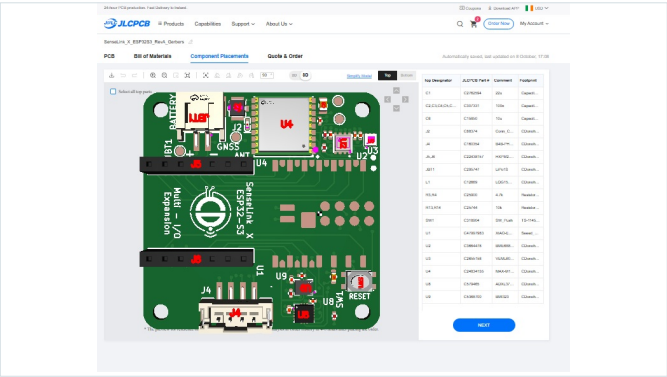
Date:	31/08/2026	Warnings:	0
Time:	17:36:45	Rule Violations:	0
Elapsed Time:	00:00:03		
Filename:	C:\Users\caoil\Git repos\STM32WL-LoRa-Sensor-Board-Portfolio\Hardware\Altium\STM32WL_RevA.PcbDoc		

Summary

Warnings	Count
Total	0

Rule Violations	Count
Clearance Constraint (Gap=0.15mm) (All),(All)	0

STM32WL Altium DRC snapshot, dated 31 Aug 2026: 0 warnings and 0 violations.



ESP32-S3 JLCPCB placement preview supporting CPL and rotation review.

Manufacturing preparation

Sourced BOMs, Gerber/drill data, component positions and assembly previews support supplier review and engineering handoff. Nordic and ESP32-S3 quotation-stage evidence is distinct from an order or fabricated board.

Model and measurement boundary: simulation and DRC support design decisions. STM32WL power, ADC and RF performance still require physical verification. No post-filter SNR improvement figure is claimed from the uncorrected calculation in the source report.

Sources: STM32WLE5 Rev 2.0, sections 6-10 and figures 11/13/22; ESP32-S3 portfolio page 7.

CAPABILITY MAP AND DETAILED EVIDENCE

Skills demonstrated across the projects

Each capability is tied to a concrete design, analysis or development-hardware example. Detailed reports are linked below.

Capability	Evidence to inspect
Altium Designer	STM32WL schematics, copper/planes and DRC; USB stack-up, impedance and routing-rule screenshots
Analogue and mixed-signal design	STM32WL protected input, buffering, anti-alias filtering, precision reference and external ADC
RF and positioning interfaces	STM32WL matching/filtering and via fencing; GNSS antenna bias; cellular antenna and SIM interfaces
Power architecture	STM32WL analogue/digital supplies; Nordic buck-boost; EdgeLink charger, power path and modem rail
Circuit and signal analysis	LTspice AC/transient and RF models; Python/Jupyter Bode, time-domain and FFT comparisons
Embedded C and Zephyr/NCS	Nordic modular firmware, sensor integration, logging, settings and custom BLE services
Embedded C++ and Wi-Fi	ESP32-S3 PlatformIO/Arduino firmware, JSON/HTTP interface and live telemetry
PCB release and handoff	Gerber/drill, BOM, CPL, placement review, source CAD and manufacturing documentation
Physical application integration	Nordic BLE and ESP32-S3 Wi-Fi sensor-to-client demonstrations on development hardware

Tool emphasis: Altium Designer for STM32WL and USB/Ethernet. KiCad for Nordic, ESP32-S3 and EdgeLink. Circuit analysis uses LTspice and Python/Jupyter; firmware uses Zephyr/NCS and PlatformIO.

Detailed project evidence

STM32WLE5 hardware design summary

USB3 to Gigabit Ethernet repository

Nordic BLE engineering portfolio v14

ESP32-S3 wireless and embedded portfolio

EdgeLink S3 Cellular hardware portfolio